

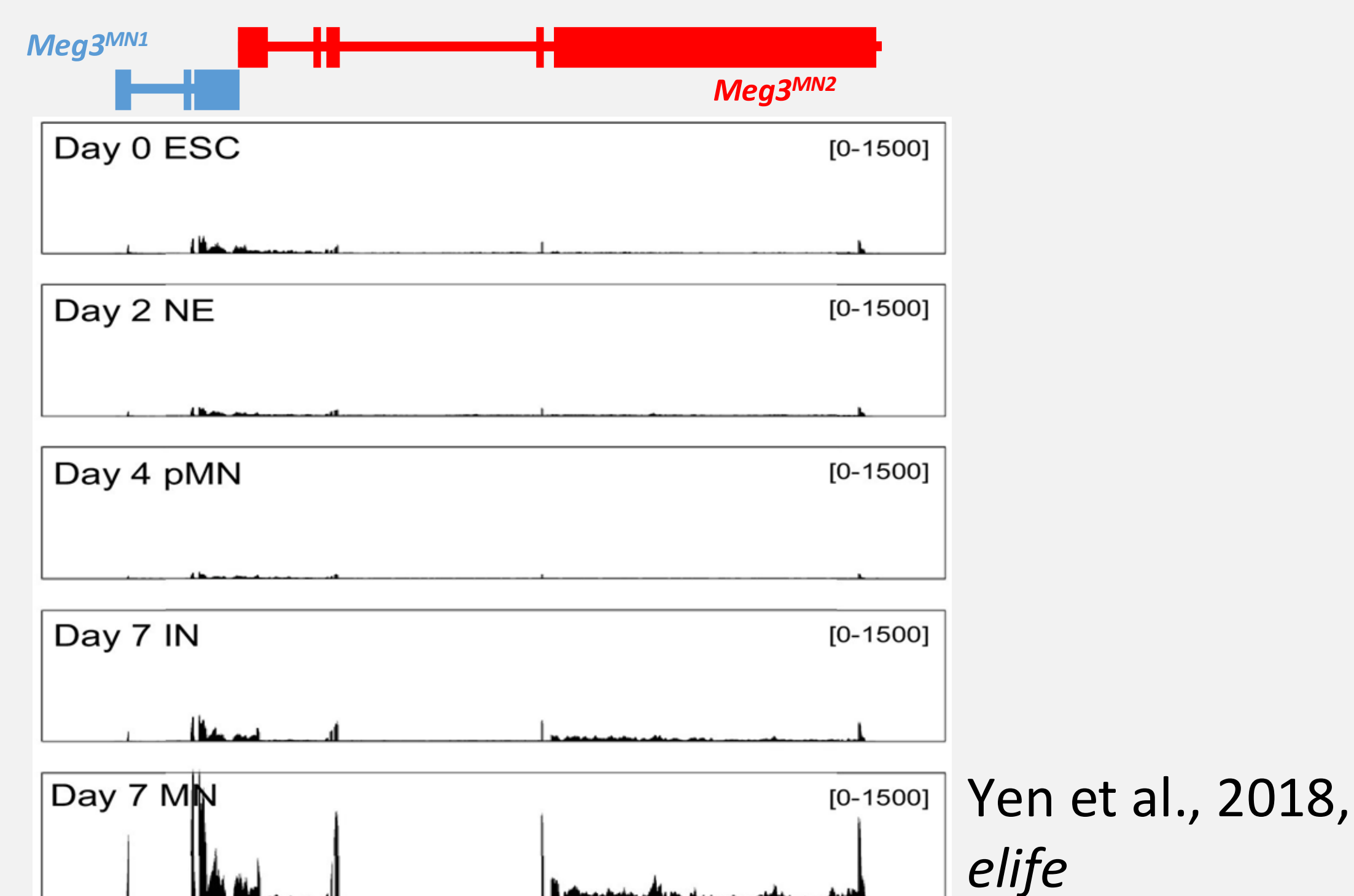
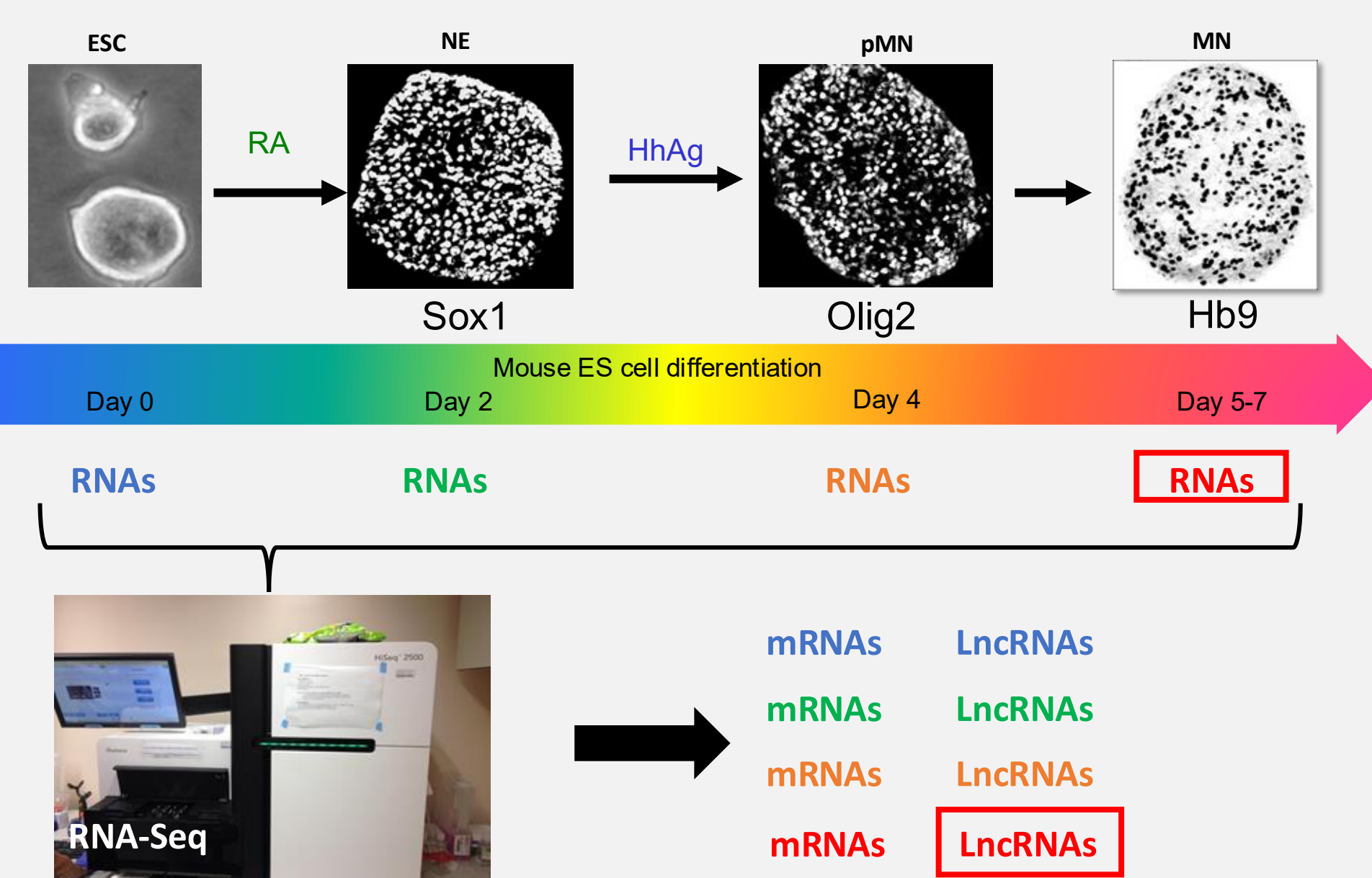
Investigation of *Meg3*-ELAVL2 mediated alternative splicing

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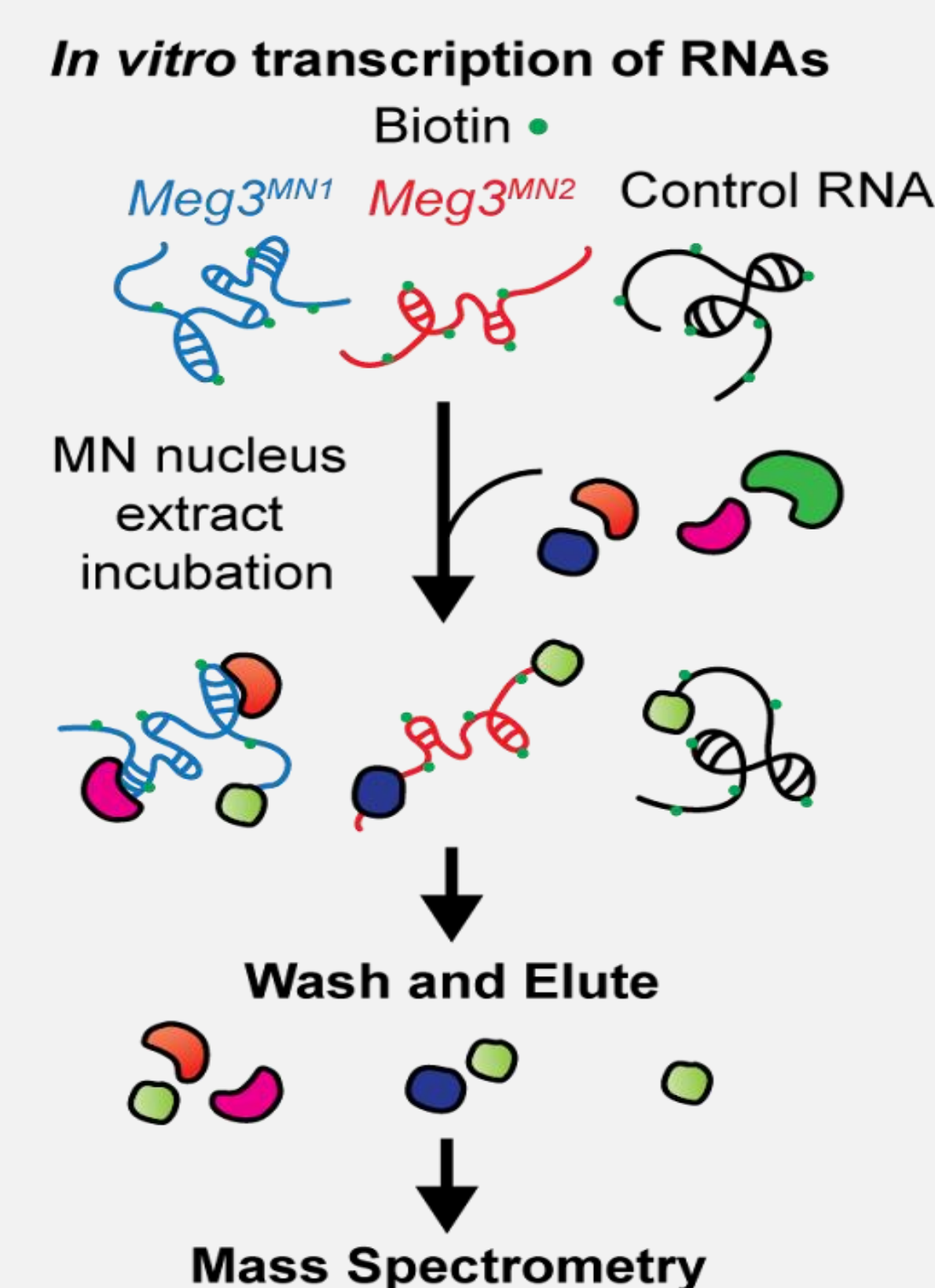
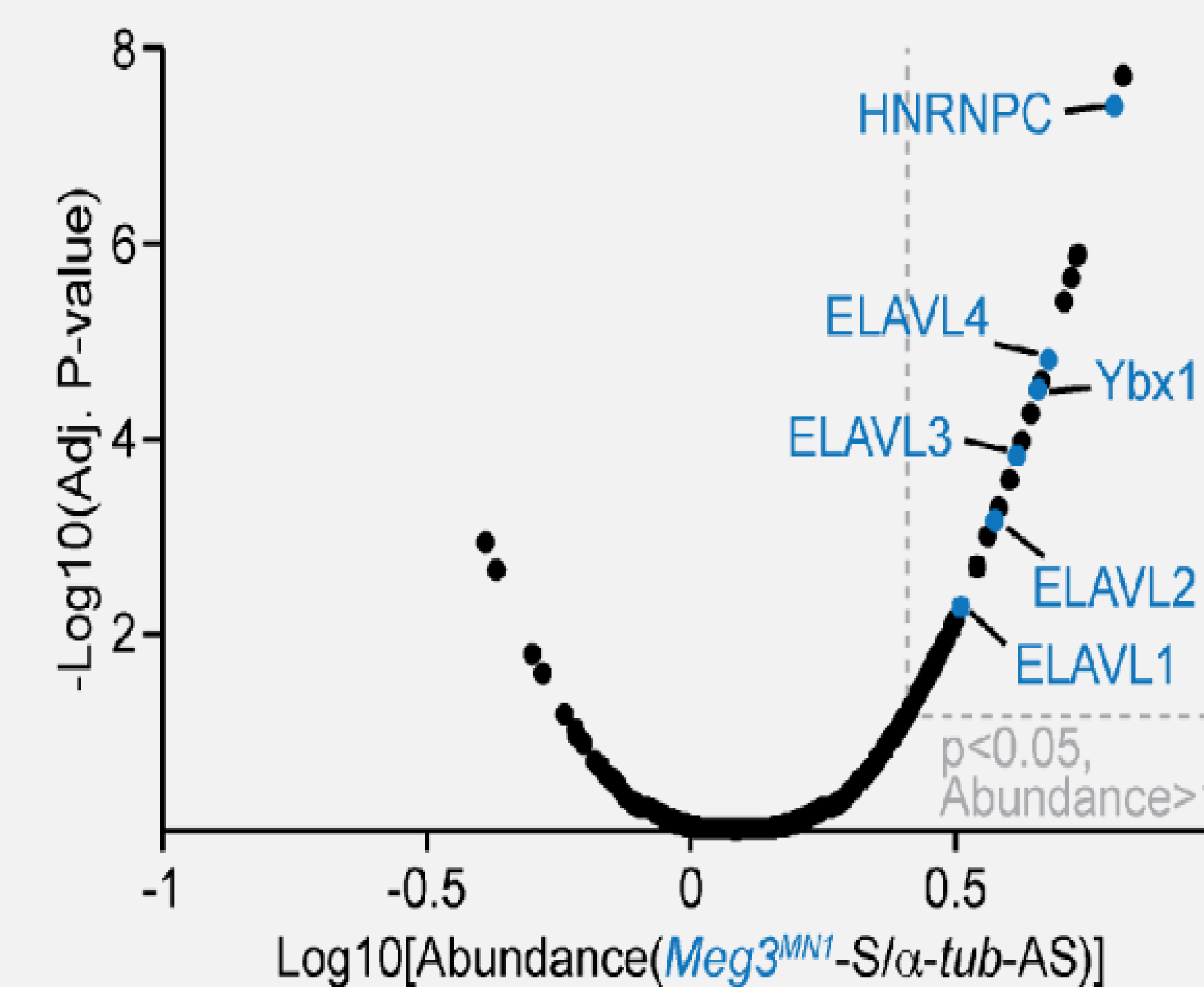
Abstract

Long non-coding RNAs have been shown to function through proteins. Our previous study shows that *Meg3*, a long non-coding RNA, is highly expressed in neurons and contains isoforms named MN1 and MN2. To understand how *Meg3* functions through proteins, we identified several proteins, including ELAVLs, that interact with *Meg3*^{MN1} using in vitro RNA pulldown assay followed by mass spectrometry. ELAVL is an RNA binding protein and has shown to regulate alternative splicing of genes. Thus, I want to determine whether *Meg3* affects ELAVL mediated splicing, and which region in ELAVL is responded to. I found the interaction between ELAVL2 and *Meg3*^{MN1} promotes splicing. Moreover, RNA recognition motif 2 (RRM2) of ELAVL2 seems to regulate *Meg3*^{MN1}-ELAVL2 mediated splicing. Together, our finding suggest a novel function of *Meg3* and may bring new insights on the regulation of splicing.

Introduction

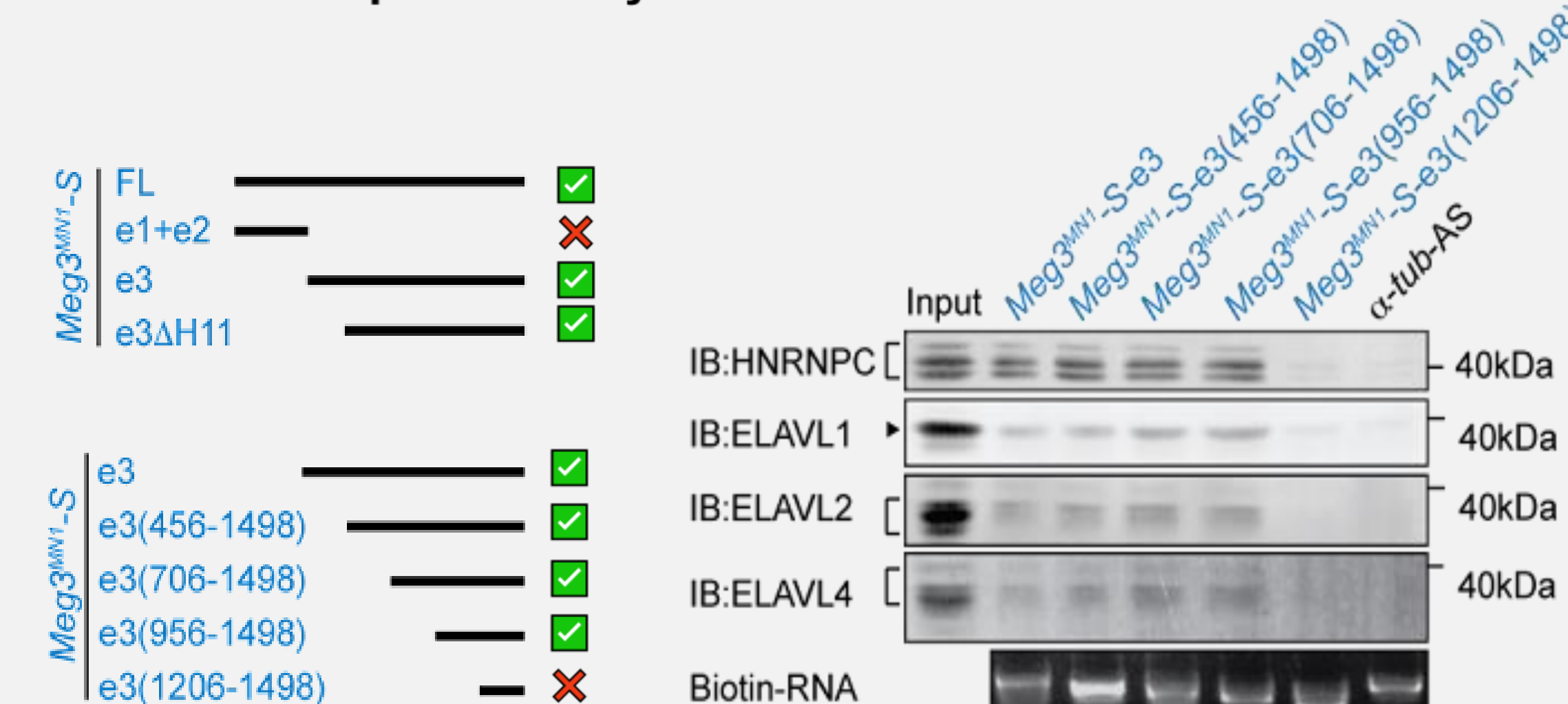


Preliminary data



Aim

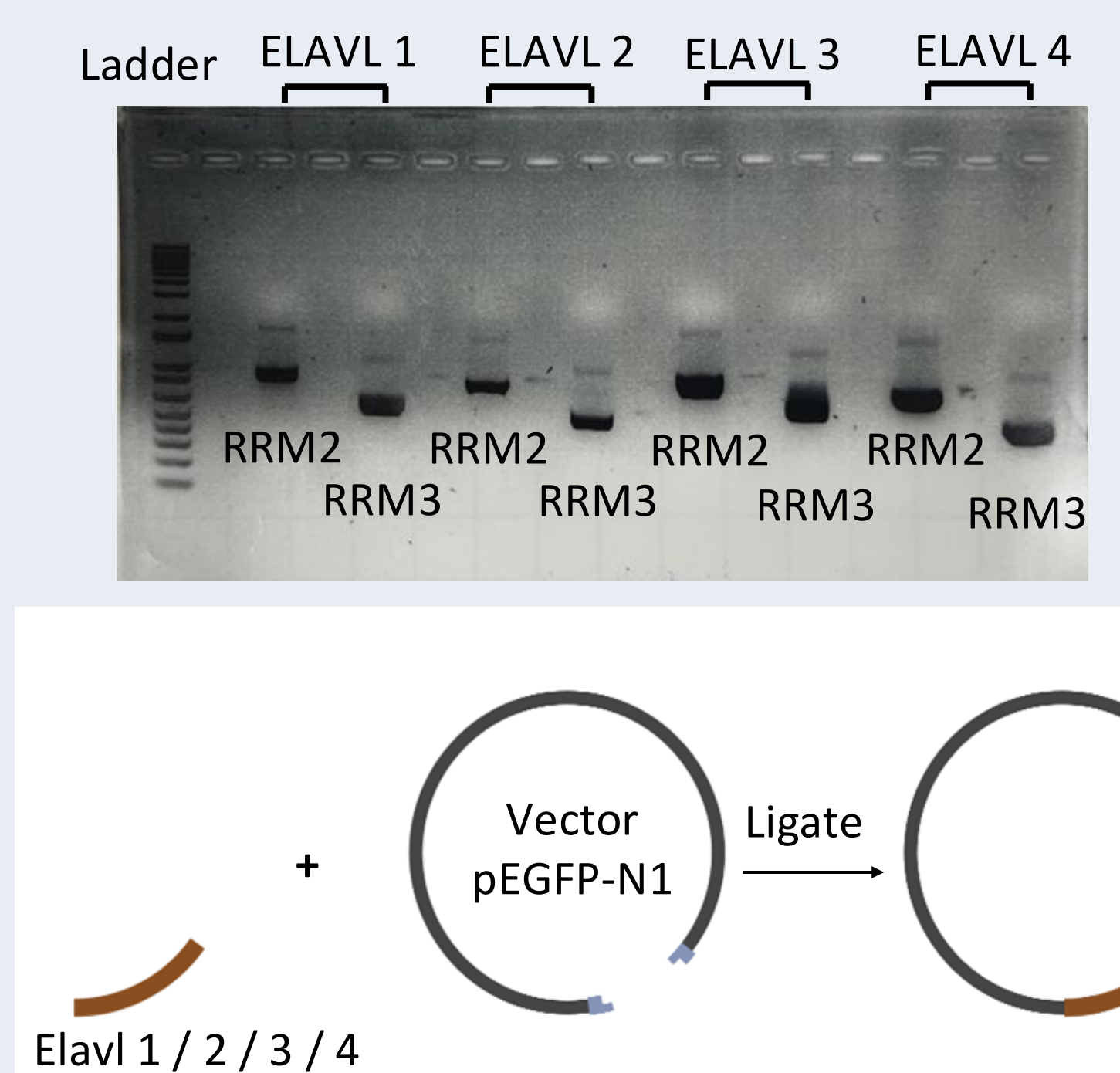
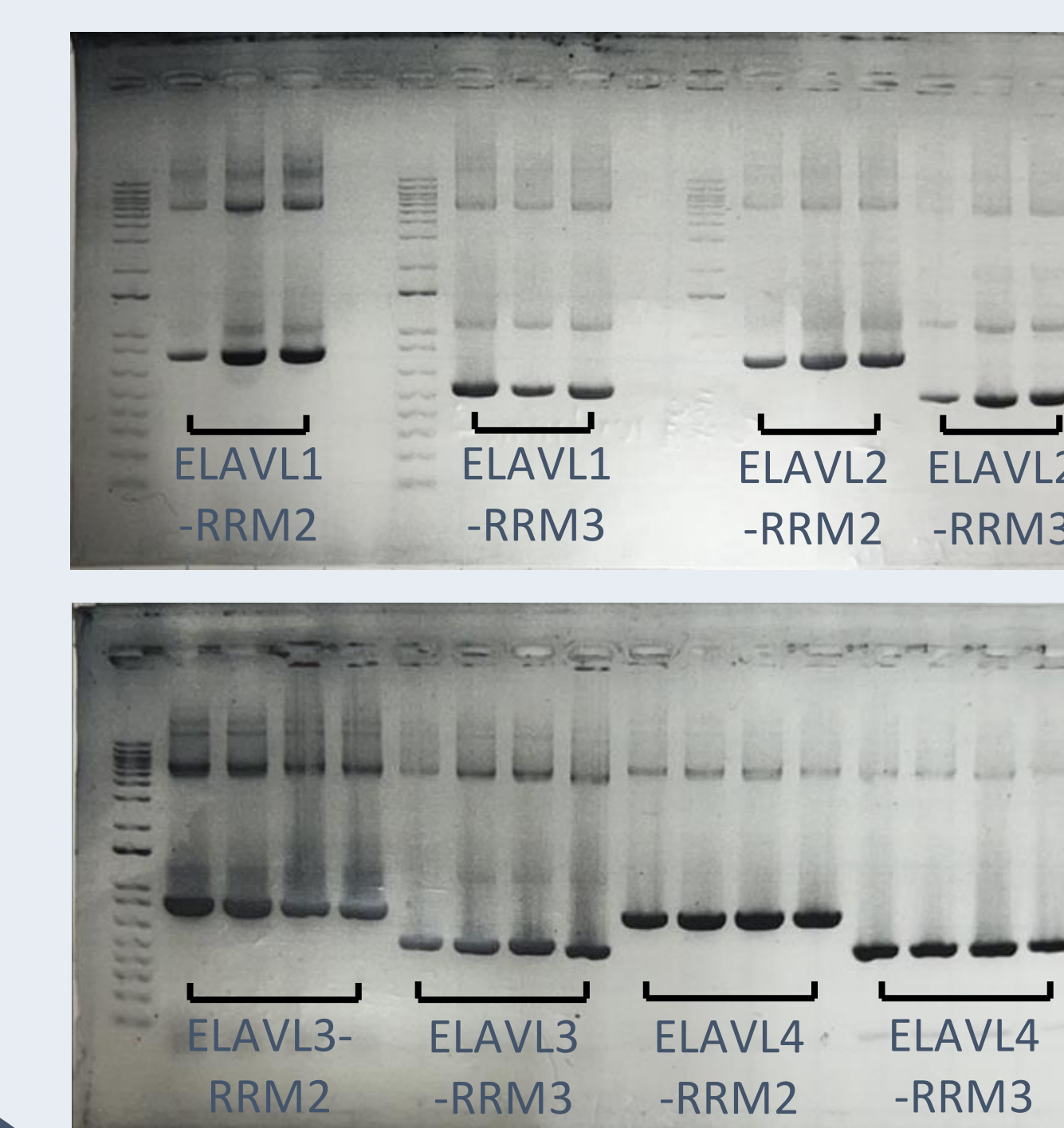
Whether *Meg3*-ELAVL2 affects alternative splicing



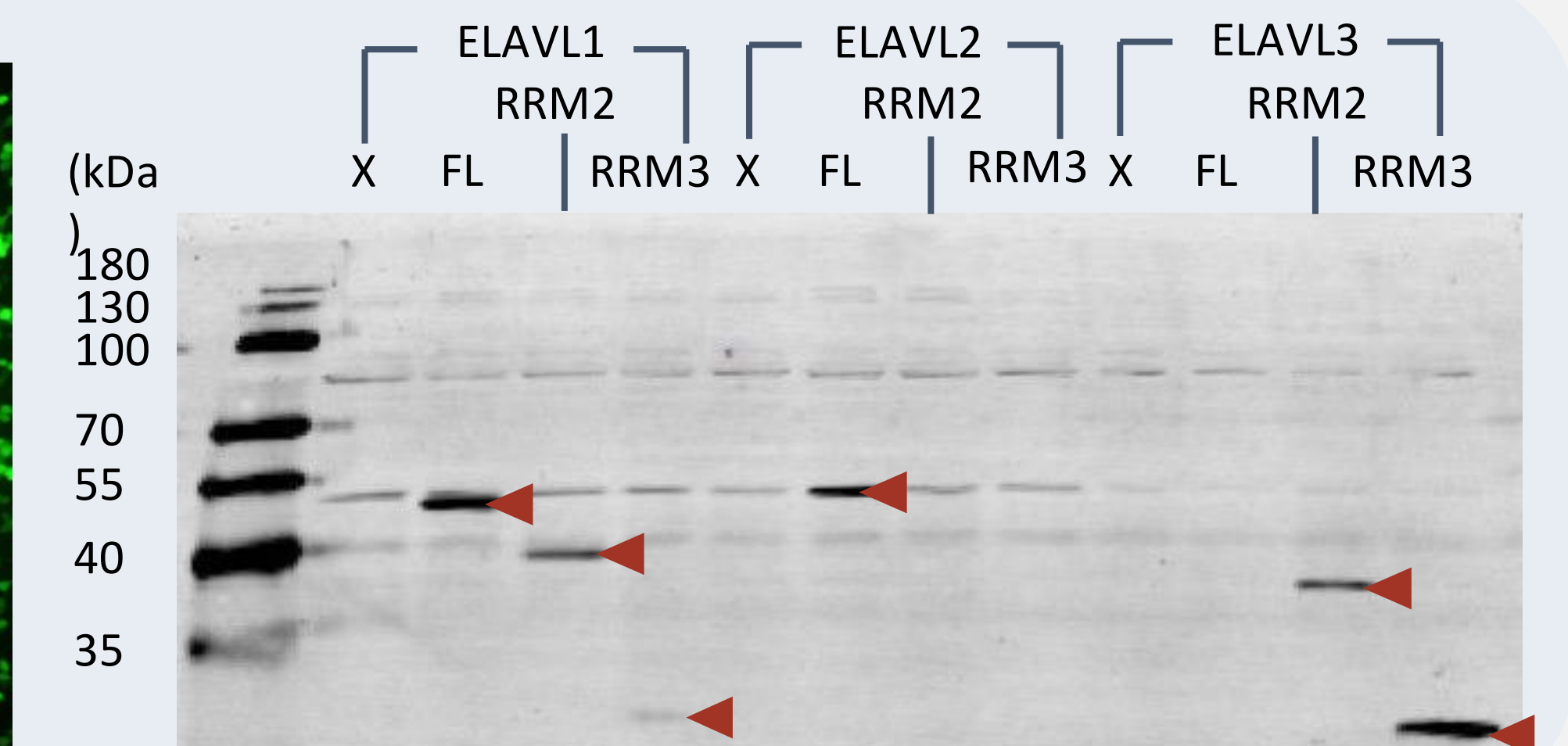
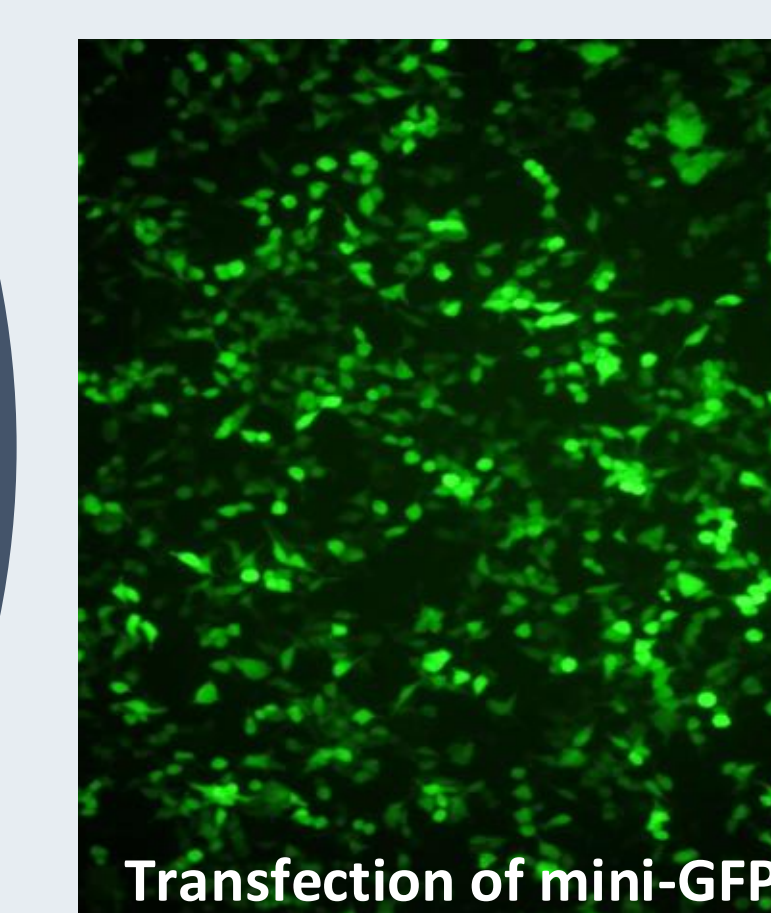
Region between 956-1206 nt of exon 3 interacts with *Meg3*^{MN1} with ELAVLs and HNRNPC

Results

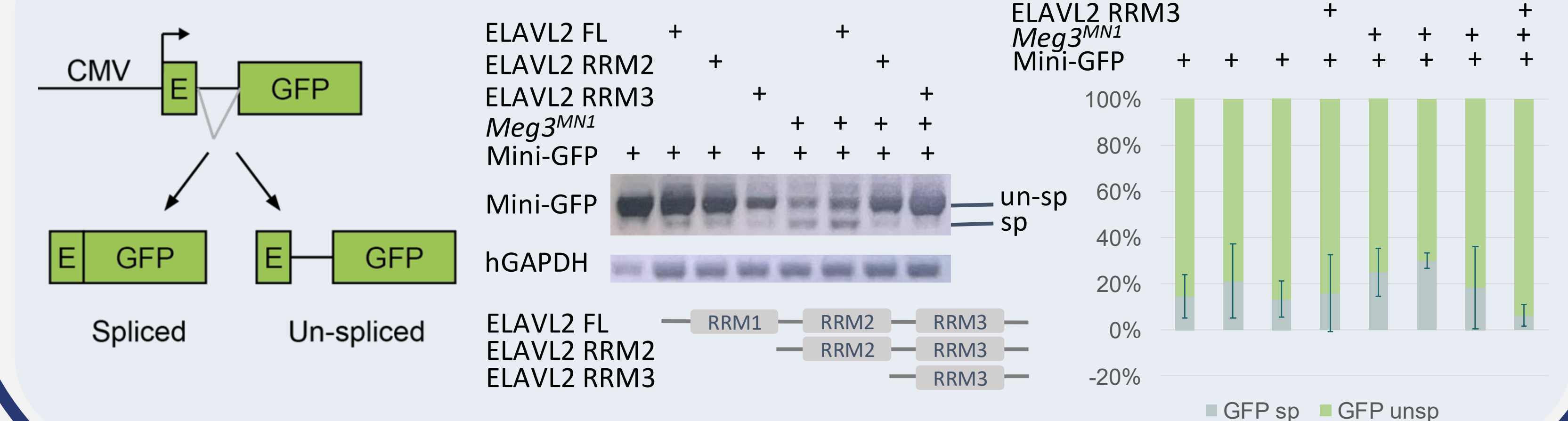
Cloning



Western Blot



Splicing



Acknowledgements

Mini-GFP gene (Bonano et al., 2007) was a kindly gift from Dr. Chien-Ling Lin from Institute of molecular biology, Academia Sinica, Taiwan